promises to construct the flyover in 450 days or less and agree to pay a penalty of ₹10,000 for each labour day spent in excess of 450 days. What is the probability that

(i) the company pays a penalty of atleast ₹2,00,000?

(ii) the company takes at most 500 days to complete the flyover?

Exercise 7.4

Choose the correct Answer

1. Normal distribution was invented by

   (a) Laplace (b) De-Moivre

   (c) Gauss (d) all the above

2. If X ~N(9,81) the standard normal variate Z will be

   (a) $Z = \frac{X-81}{9}$ (b) $Z = \frac{X-9}{81}$

   (c) $Z = \frac{X-9}{9}$ (d) $Z = \frac{9-X}{9}$

3. If Z is a standard normal variate, the proportion of items lying between $Z = -0.5$ and $Z = -3.0$ is

   (a) 0.4987 (b) 0.1915

   (c) 0.3072 (d) 0.3098

4. If X ~N(μ, σ^2), the maximum probability at the point of inflexion of normal distribution is

   (a) $\left(\frac{1}{\sqrt{2\pi}}\right)e^{\frac{1}{2}}$

   (b) $\left(\frac{1}{\sqrt{2\pi}}\right)e^{\left(-\frac{1}{2}\right)}$

   (c) $\left(\frac{1}{\sigma\sqrt{2\pi}}\right)e^{\left(-\frac{1}{2}\right)}$

   (d) $\left(\frac{1}{\sqrt{2\pi}}\right)$

5. In a parametric distribution the mean is equal to variance is

   (a) binomial (b) normal

   (c) poisson (d) all the above

6. In turning out certain toys in a manufacturing company, the average number of defectives is 1%. The probability that the sample of 100 toys there will be 3 defectives is

   (a) 0.0613 (b) 0.613

   (c) 0.00613 (d) 0.3913

7. The parameters of the normal distribution

   $f(x) = \left(\frac{1}{\sqrt{72\pi}}\right)\frac{e^{-(x-10)^2}}{72}$ $-\infty < x < \infty$

   (a) (10,6) (b) (10,36)

   (c) (6,10) (d) (36,10)

8. A manufacturer produces switches and experiences that 2 per cent switches are defective. The probability that in a box of 50 switches, there are atmost two defective is :

   (a) 2.5 e^{-1}

   (b) e^{-1}

   (c) 2 e^{-1}

   (d) none of the above

   RGECXV

9. An experiment succeeds twice as often as it fails. The chance that in the next six trials, there shall be at least four successes is

   (a) 240/729 (b) 489/729

   (c) 496/729 (d) 251/729

10. If for a binomial distribution B(n,p) mean = 4 and variance = 4/3, the probability, $P(X \geq 5)$ is equal to

    (a) $(2/3)^6$ (b) $(2/3)^5(1/3)$

    (c) $(1/3)^6$ (d) $4(2/3)^6$

XII Std - Business Maths & Stat EM Chapter 7.indd 169 2/29/2020 11:34:40 AM